

Name: Ineza Serucaca Ines **Student ID:** 20251SEN290

Overview

I analysed 60 days of Kigali weather (1 September to 30 October 2026) using NumPy arrays and boolean masking. The dataset covers temperature, rainfall and humidity. Temperatures averaged 24.00 °C (standard deviation 2.61 °C), ranging from 19.3 °C to 28.6 °C. The hottest day was 17 September at 28.6 °C.

Main insight: the period cooled by about 4.4 °C

The clearest pattern is a seasonal shift. The first 30 days averaged 26.21 °C, while the last 30 averaged 21.78 °C, a drop of about 4.4 °C. The anomaly analysis supports this. Nine days were more than 3 °C above the 24.00 °C mean (7, 9, 11, 12, 14, 16, 17, 19 and 22 September), and all fell in the first half of the period. Another 8 days were more than 3 °C below the mean. Together these show that the hottest weather came early and the cooler weather came later.

Rain did not explain the cooling

Rain was frequent but not strongly linked to temperature. There were 16 rainy days with 331.2 mm in total, and the heaviest single day was 26.2 mm on 7 September. Rainy days averaged 24.14 °C and dry days 23.95 °C, a gap of only about 0.2 °C. Five days were both hot (≥ 26 °C) and rainy. This suggests the cooling comes from the changing season rather than from individual rain events.

Other observations

Rainfall was slightly lower in the second half (148.0 mm) than in the first (183.2 mm), while average humidity rose from 58.1% to 66.5%. Sixteen days had humidity of 70% or more. These two changes together with the temperature drop are consistent with a seasonal transition, though 60 days of data from a single year is too little to draw firm conclusions.

Method

All calculations used vectorised NumPy operations (mean, argmax, boolean masks, slicing) with no loops, apart from printing the bonus ranking. Masks made it easy to compare subsets of the data, such as hot days or rainy days, in a single line each.